Kevin D. Silva-Pérez

PhD Student in Computational Mathematics and Statistics
University of Puerto Rico – Río Piedras

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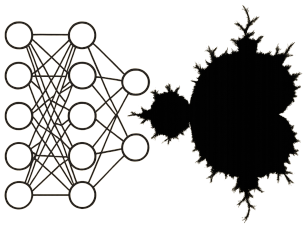
ResearchGate | GitHub

Research Interests

Partial differential equations (PDEs); analysis on fractals and ramified domains; diffusion processes and heat flow in complex geometries; fractional Laplacian; functional and classical analysis; applications of PDEs to biological and biomedical systems; connections between PDEs and machine learning / operator learning.

Current Research Projects (In Progress)

- Fractional Laplacian operators on non-smooth and irregular sets with applications to anomalous diffusion.
- PDE-informed machine learning, including DeepONets and PINNs for scientific computing.



Education

University of Puerto Rico, Río Piedras 2023–Present
PhD in Computational Mathematics and Statistics

University of Puerto Rico, Mayagüez 2021–2023
MSc in Applied Mathematics

Universidad de Cartagena, Colombia 2017–2020
Bachelor in Mathematics

Advisor: Alejandro Vélez-Santiago, PhD
Master Thesis: Diffusion over Domains with Ramified Fractal Boundaries
Book editor: *Laboratory in Differential Calculus with MATLAB*

Research Experience

Researcher (under funded project) Fall 2025 – Present
University of Puerto Rico – Río Piedras
Funded project: FIPI 2024–2026, Award Number: 20FIPI570025
New trends in regularity theory for boundary value problems over non-smooth domains
Principal Investigator: Dr. Alejandro Vélez-Santiago
Role: Theoretical analysis of boundary value problems on irregular domains and regularity methods

Research Assistant (under funded project) Fall 2022 – Summer 2023
Puerto Rico Science, Technology & Research Trust
Funded project: Agreement 2022-00014
Boundary value problems of nonstandard growth structure over real-world regions
Principal Investigator: Dr. Alejandro Vélez-Santiago (UPR–Mayagüez)
Role: Mathematical modeling and analysis of PDEs with nonstandard growth

Academic Positions

University of Puerto Rico – Río Piedras

- **Researcher (under funded project)** Fall 2025 – Present
  FIPI 2024–2026 Project: New trends in regularity theory for boundary value problems over non-smooth domains
  Principal Investigator: Dr. Alejandro Vélez-Santiago
- **Teaching Assistant** Fall 2023 – Summer 2025
  Courses: MATE 3105 (Mathematical Appreciation), MATE 3171–3172 (Precalculus I–II), MATE 3151 (Calculus I)

University of Puerto Rico – Mayagüez

- **Research Assistant** Fall 2022 – Summer 2023
  Boundary value problems of nonstandard growth structure over real-world regions
  PI: Dr. Alejandro Vélez-Santiago
- **Teaching Assistant / Instructor** Fall 2021 – Summer 2022
  Precalculus I–II, Calculus I–III
- **College Board and SAT Instructor** Sep 2021 – Dec 2022

Tutor, Universidad de Cartagena, Colombia Aug 2019 – May 2021
Tutoring in Calculus I–III, Differential Equations, and Statistics.

Publications

Research Articles

- Silva-Pérez, K., Vélez-Santiago, A. Diffusion over ramified domains: solvability and fine regularity. *Calculus of Variations and Partial Differential Equations* **64** (2025), Article no. 238. https://link.springer.com/article/10.1007/s00526-025-03103-5

Books

- Silva-Pérez, K., Arrieta-Montalvo, R., Ortega, P., Johnson, B. *Calculus with MATLAB*. Editorial Universidad de Cartagena, April 2021.

Selected Talks and Conferences

Great Plains Operator Theory Symposium (GPOTS 2025), University of Puerto Rico – Río Piedras October 20–24, 2025
Talk: *Elliptic Problem over Ramified Domains.*

Workshop in Partial Differential Equations: Theory and Applications, University of Puerto Rico — Mayagüez, PR October 7–10, 2025
Poster: *Diffusion over Bronchial Trees: Construction and Approximation Results.*

Inaugural Talk, TURI Student Association, University of Puerto Rico — Río Piedras September 17, 2025
Talk: *Huellas fractales en el universo.*

LatMath 2025, IPAM, UCLA, Los Angeles, CA March 7–11, 2025
Poster: *Diffusion on Bronchial Trees: Construction and Approximation Results.*

Café y Ciencia, Universidad de Cartagena, Colombia June 10, 2024
Talk: *PDEs & Deep Learning: Toward a New Paradigm in Operator Learning.*

Workshop in Non-Commutative Geometry and Higher Index Theory, Luquillo, PR April 5–7, 2024

Analysis on Fractals and Networks, and Applications, CIRM, France March 18–22, 2024
Poster: *Diffusion on Bronchial Trees: Construction and Approximation Results.*

XXXIX Interuniversity Seminar on Research in the Mathematical Sciences (SIDIM), Humacao, PR March 1–2, 2024

Café y Ciencia, Universidad de Cartagena, Colombia June 20, 2023
Talk: *Hausdorff Measure over Domains with Ramified Fractal Boundaries.*

13th AIMS Conference on Dynamical Systems, Differential Equations and Applications, Wilmington, NC May 31 – June 4, 2023
Talk: *Diffusion over Bronchial Trees: Fine Regularity* (with A. Vélez-Santiago).

XXXVIII Interuniversity Seminar on Research in the Mathematical Sciences (SIDIM), Mayagüez, PR Feb 24–25, 2023
Talk: *Diffusion on Bronchial Trees: Construction and Approximation Results.*

Latinx in the Mathematical Sciences Conference, UCLA July 7–9, 2022

7th Cornell Conference on Analysis, Probability, and Mathematical Physics on Fractals, Cornell University, NY June 4–8, 2022

Teaching and Skills

Teaching interests: Calculus (single and multivariable), Differential Equations, Linear Algebra, Real and Functional Analysis, Introductory Probability and Statistics, PDEs and mathematical modeling.

Technical skills: Python, MATLAB, R, LaTeX.

Languages: Spanish (native), English (professional working proficiency).

Academic Service and Memberships

Reviewer, Colciencias (Colombia) 2022–2024
Evaluation of scientific articles and research proposals in mathematics and related areas.

Member, SACNAS (Society for Advancement of Chicanos/Hispanics and Native Americans in Science).
Member, Asociación Estudiantil de Matemáticas, UPRRP.
Volunteer, American Red Cross, Puerto Rico (Information and Planning Team).
Member, Fractals and PDEs Research Group, directed by Alejandro Vélez-Santiago, PhD.

Leadership & Initiatives

Founder and Academic Advisor – TURI (Teoría, Universo, Razonamiento e Investigación), University of Puerto Rico – Río Piedras 2024–Present

Interdisciplinary academic initiative focused on mathematics, AI, physics, research workshops, and scientific communication.